

RuDocGround-CF: A Counterfactual Multi-Document Benchmark for Russian LLM Evaluation

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Abstract

Russian-language LLM evaluation increasingly requires operational tests that combine authority, time, evidence, and machine-readable decisions across document bundles. We introduce RuDocGround-CF, a benchmark that complements MERA with controlled corporate-style multi-document reasoning. The v1.3.1 release contains 30 cases, 80 instantiated variants, 1,160 tasks, and 556 labeled cross-variant relations. Each case evaluates exact answers, operational decisions, evidence grounding, missing-information handling, structured-output reliability, and responses to material or nuisance document edits. On a locally served Qwen3.5 4B baseline, independent single-question inference achieves 0.2845 strict answer accuracy (95% case-cluster bootstrap CI [0.2234, 0.3427]), while variant-batched compact inference reaches 0.7586. The paired protocol gap is 0.4741 [0.4323, 0.5162] and is positive in all 10,000 bootstrap replicates. Beyond 239 terminal format failures, 290 parser-accepted boolean responses omit the canonical answer field. RuDocGround-CF adds protocol-sensitive, counterfactual, and interface-aware evaluation to the Russian LLM benchmark landscape.

Keywords: Russian NLP; multi-document reasoning; counterfactual evaluation; grounded question answering; structured outputs; protocol sensitivity

1. Introduction

Consider a travel-expense case in which a general policy sets a lodging limit of RUB 9,000 per night, while a temporary order raises the limit to RUB 12,000. A one-line change to the order's expiration date determines whether the fourth hotel night remains covered by the temporary rule. The maximum allowable amount changes from RUB 48,000 to RUB 45,000, yet a question about the baseline policy must still return RUB 9,000. Solving the case requires document authority, temporal scope, arithmetic, and selective sensitivity to the edited record.

Corporate document workflows routinely combine policies, appendices, orders, approvals, invoices, correspondence, and system extracts. A useful model must identify the controlling record, ground its conclusion, detect missing prerequisites, and respond correctly when one document changes. Aggregate answer accuracy hides several operational failures: unsupported evidence, stale conclusions after an amendment, overreaction to an irrelevant edit, and malformed outputs that cannot enter a downstream workflow.

RuDocGround-CF evaluates these behaviors through controlled multi-document variants. Variant A provides the baseline case, variant B changes a causally relevant fact, and variant C changes a nuisance or secondary detail. Each question is labeled with an expected answer and, where applicable, an expected cross-variant relation. The output contract separates canonical answers, operational decisions, evidence, missing-information codes, and explanations.

RuDocGround-CF reveals pronounced protocol sensitivity. The same frozen model-data pair yields 28.45% strict answer accuracy under independent single-question inference and 75.86% under compact variant-batched inference. The paired case-cluster gap is 47.41 percentage points [43.23, 51.62]. We define the difference between these complete inference settings as the protocol effect.

This paper makes three contributions:

- RuDocGround-CF, a Russian-language benchmark for controlled corporate-style multi-document evaluation with 30 cases, 80 variants, 1,160 tasks, evidence annotations, and 556 labeled counterfactual relations.
- An evaluation framework that separates strict end-to-end reliability, conditional relation behavior, and joint endpoint correctness.
- A reproducible Qwen3.5 4B baseline with a statistically stable protocol gap, systematic boolean-answer omission, and measurable selection effects in conditional counterfactual reporting.

2. Related Work

2.1 Russian-language and document-level evaluation

SuperGLUE, RussianSuperGLUE, and MERA provide broad evaluations across general language understanding, Russian inference, commonsense, and instruction-following tasks [1-3]. Multi-document and document-level benchmarks contribute

complementary capabilities: HotpotQA and MuSiQue test evidence integration [4, 11], Qasper anchors questions in full scientific papers [5], ContractNLI combines contract inference with evidence spans [6], and LegalBench covers a wide range of legal reasoning tasks [7]. RuDocGround-CF complements MERA with controlled corporate multi-document reasoning, document-grounded evidence, structured operational outputs, and explicit cross-variant relations.

2.2 Long context, behavioral tests, and counterfactuals

LongBench, Lost in the Middle, and RULER show that nominal context windows do not guarantee reliable use of dispersed information [8-10]. CheckList and contrast sets demonstrate how targeted perturbations reveal local decision boundaries that aggregate accuracy misses [12, 13]. CounterBench studies formal counterfactual reasoning over rule-based structures [18]. RuDocGround-CF applies this behavioral logic to document systems in which authority, time, and evidence are distributed across records.

2.3 Protocol-aware and structured-output evaluation

HELM establishes transparent multi-metric evaluation with released prompts and outputs [14], while prompt-order studies demonstrate that presentation choices materially alter model behavior [15]. Structured-output research documents substantial variation in JSON validity and schema compliance across models and tasks [16, 17]. RuDocGround-CF evaluates content and interface reliability in one benchmark: malformed and noncanonical responses remain failures in strict end-to-end scoring.

3. RuDocGround-CF Benchmark

3.1 Design and composition

The benchmark enforces five design principles. Every case requires multiple documents; evidence resolves to document identifiers and locators; controlled edits separate material changes from nuisance changes; answers use canonical machine-checkable types; and strict evaluation retains schema failures as part of system reliability. Version 1.3.1 contains 30 controlled Russian-language cases spanning travel, procurement, approvals, amendments, authority, finance, human resources, privacy, information security, intellectual property, logistics, property, reconciliation, shipping, and notices.

Every case includes the A/B material-intervention pair, and 20 cases also include variant C for a designated nuisance intervention, yielding 80 instantiated variants.

Table 1. Benchmark composition.

Property	Value
Cases	30
Instantiated variants	80
Prompt / gold rows	1,160 / 1,160
Canonical answer types	11
Counterfactual relation instances	556
Language	Russian
Setting	Corporate-style multi-document reasoning

3.2 Running example

A simplified view of trip_001 illustrates the construction. A policy provides the baseline limit, a temporary order changes the operative limit, and a hotel invoice records four nights. Variant B changes only the order’s end date. The edited date changes calculations that depend on the fourth night while leaving the baseline-policy question unchanged.

Box 1. Simplified A/B intervention from a travel-expense case.

Element	Variant A	Variant B	Expected relation
Temporary order	RUB 12,000 applies to all four nights	Order expires before the fourth night	Material edit
Baseline policy question	RUB 9,000	RUB 9,000	Invariant
Maximum allowable lodging	RUB 48,000	RUB 45,000	Flip
Reimbursable lodging	RUB 45,600	RUB 43,200	Flip

Variant C changes a lower-level nuisance detail. Questions about that detail update, while higher-level conclusions remain stable.

3.3 Controlled relation families

Each labeled pair belongs to one of four namespaces. A-B flip and A-B invariant measure sensitivity and stability around the material intervention. A-C nuisance flip asks whether the model notices the changed nuisance detail. A-C causal invariant checks whether the higher-level conclusion survives that secondary edit.

Table 2. Counterfactual relation families.

Relation	Expected behavior	Count
A-B flip	Material edit changes the answer	99
A-B invariant	Unrelated answer remains stable	181
A-C nuisance flip	Changed nuisance detail is detected	55
A-C causal invariant	Higher-level conclusion remains stable	221

3.4 Task schema and answer types

Every task is keyed by case, variant, and question identifiers. The independent protocol requests six fields: answer, decision, evidence, evidence details, missing information, and explanation. Answers are canonicalized as boolean, categorical, code set, date, date range, datetime, identifier, integer, money, status, or threshold values. Decision labels appear only where an operational judgment is required. Evidence uses document identifiers and optional locators; missing information uses canonical codes. Explanations support audit but are not exact-match targets.

3.5 Gold construction and release integrity

Gold annotations were derived from the controlling documents and the designed A/B/C interventions. Each task received a canonical answer, an operational decision where applicable, required and supporting evidence, missing-information codes, and an expected cross-variant relation. Calculated answers were checked against source values and intervention logic before inclusion in the frozen release. The release contains 1,160 prompt keys and 1,160 aligned gold keys, with no missing or orphan records. Validation covers schema conformance, key alignment, expected relation denominators, leakage checks, and a perfect-prediction fixture that scores 1.0 throughout. Content hashes identify the benchmark, predictions, reports, and statistical artifacts.

4. Experimental Setup

4.1 Model and environment

We evaluate qwen3.5:4b through Ollama 0.30.10. The recorded model identifier is 2a654d98e6fb. Runs were executed on a MacBook Air with an Apple M4 processor and 16 GB of unified memory. The complete benchmark runs on consumer hardware. The release records the model identifier, serving version, hardware, prompts, predictions, reports, and content hashes.

4.2 Evaluation protocols

The study compares two complete inference protocols. Independent single-question inference sends one document variant, one question, and a full task-specific JSON schema per request, for 1,160 calls. Variant-batched compact inference sends one document variant together with all associated questions, for 80 calls. The compact response uses short fields for answer, decision, evidence, and missing information; local document identifiers are then expanded deterministically.

The two protocols define distinct inference settings through question visibility, schema compactness, and call structure. Their measured difference is the protocol effect.

Table 3. Protocol comparison.

Feature	Independent single-question	Variant-batched compact
Call unit	One question	All questions for one variant
Calls	1,160	80
Output contract	Full task-specific object	Compact shared fields
Neighboring questions	Hidden	Visible
Primary reporting	Strict end-to-end + conditional analysis	End-to-end benchmark metrics

4.3 Validation and metrics

The evaluator enforces canonical output types. Free-text numbers fail the integer contract, and JSON booleans fail string-valued decision labels. Strict end-to-end metrics retain the full applicable denominator and assign zero to invalid outputs. Conditional metrics describe semantic behavior on evaluator-retained responses. Decision accuracy is computed over 693 decision-required tasks in the strict layer and 501 evaluator-retained decision outputs in the conditional layer.

Answer accuracy is exact canonical match. Evidence precision, recall, and F1 are macro-averaged over tasks, with invalid rows scored as zero under strict evaluation. The evaluable-output rate is the fraction of tasks retained by the released validator. Missing-information exact match is reported separately.

For each pair, relation accuracy records whether predictions change or remain stable as expected. Strict relation scoring counts an invalid endpoint as failure; valid-pair scoring conditions on two evaluator-retained endpoints. We also add joint endpoint-correct relation accuracy, which requires both endpoint answers to match gold and the predicted relation to hold. This metric prevents identical wrong answers from receiving full credit for invariance.

4.4 Statistical analysis

Rows from the same case share documents, questions, and perturbations. We therefore estimate uncertainty with a case-clustered bootstrap rather than resampling 1,160 rows independently. Each of 10,000 replicates samples 30 case identifiers with replacement and includes every task and relation attached to each sampled case. The seed is 42 and reported intervals are percentile 95% confidence intervals. Protocol differences use paired case resampling.

5. Results

5.1 Independent strict end-to-end performance

Independent inference yields 921 evaluator-retained records from 1,160 tasks. Strict answer accuracy is 0.2845, decision accuracy is 0.4834, and evidence F1 is 0.4254. The evaluable-output rate is 0.7940. Case-cluster intervals quantify performance across the 30 benchmark cases.

Table 4. Independent strict end-to-end results.

Metric	Point estimate	95% case-cluster CI
Answer accuracy	0.2845	[0.2234, 0.3427]
Decision accuracy	0.4834	[0.4295, 0.5417]

Metric	Point estimate	95% case-cluster CI
Evidence precision	0.4179	[0.3382, 0.4960]
Evidence recall	0.4650	[0.3789, 0.5474]
Evidence F1	0.4254	[0.3457, 0.5019]
Evaluable-output rate	0.7940	[0.7563, 0.8333]

5.2 Protocol sensitivity

Variant-batched compact inference reaches 0.7586 answer accuracy. The paired difference from independent strict evaluation is 0.4741 [0.4323, 0.5162], and all 10,000 bootstrap replicates are positive. Decision accuracy rises by 0.2973, evidence F1 by 0.2482, and the evaluable-output rate by 0.2060.

Table 5. Paired protocol effects.

Metric	Batched	Independent strict	Difference	95% CI
Answer accuracy	0.7586	0.2845	0.4741	[0.4323, 0.5162]
Decision accuracy	0.7807	0.4834	0.2973	[0.2075, 0.3813]
Evidence F1	0.6736	0.4254	0.2482	[0.1731, 0.3180]
Evaluable-output rate	1.0000	0.7940	0.2060	[0.1667, 0.2437]

The case-level comparison in Figure 1 shows a positive answer-accuracy gap in all 30 cases. The smallest difference is 0.1563 in asset_scope_010; the largest is 0.7619 in privacy_017. Several cases fall to zero under independent strict scoring while retaining moderate or high batched accuracy.

5.3 Counterfactual behavior and endpoint correctness

Strict relation scores range from 0.1717 for A-B flips to 0.5455 for A-C nuisance flips. Conditioning on two evaluator-retained endpoints raises all four relation scores above 0.94, but the retained subset is small: only 18 of 99 A-B flip pairs and 63 of 221 A-C causal-invariance pairs remain evaluable.

Joint endpoint-correct scoring clarifies what survives the relation test. Fourteen of 99 A-B flip pairs satisfy the expected change with both answers correct. The corresponding counts are 67/181 for A-B invariance, 30/55 for A-C nuisance flip, and 52/221 for A-C causal invariance. Among retained pairs, joint correctness is 0.778, 0.893, 1.000, and 0.825 respectively. Conditional relation consistency is therefore stronger than conditional endpoint accuracy.

Table 6. Counterfactual scoring layers.

Relation	Strict relation	Strict joint-correct	Valid-pair relation	Retained pairs
A-B flip	17/99 = 0.1717	14/99 = 0.1414	17/18 = 0.9444	18/99
A-B invariant	73/181 = 0.4033	67/181 = 0.3702	73/75 = 0.9733	75/181
A-C nuisance flip	30/55 = 0.5455	30/55 = 0.5455	30/30 = 1.0000	30/55
A-C causal invariant	61/221 = 0.2760	52/221 = 0.2353	61/63 = 0.9683	63/221

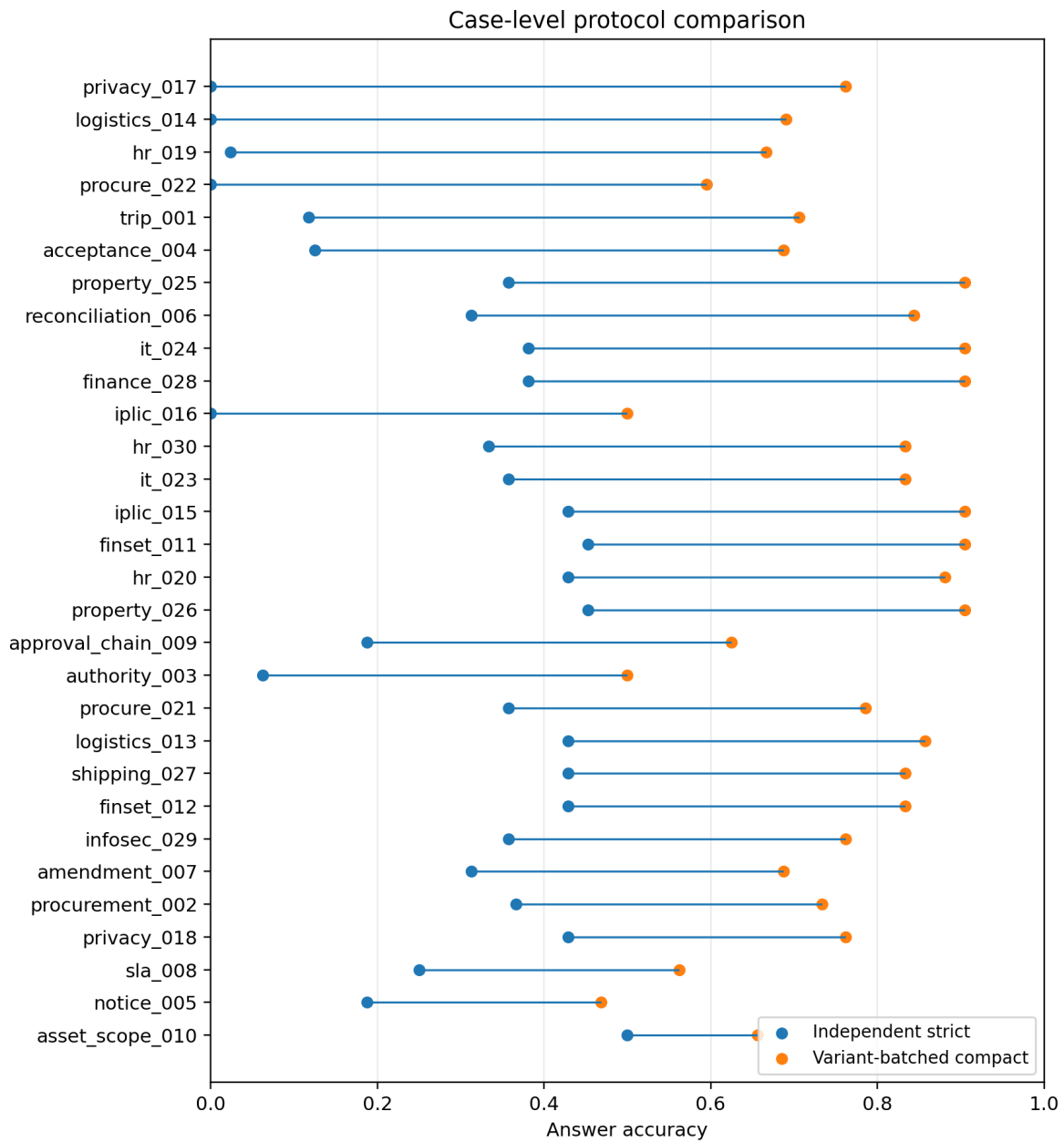


Figure 1. Case-level answer accuracy under the two protocols, sorted by the within-case gap.

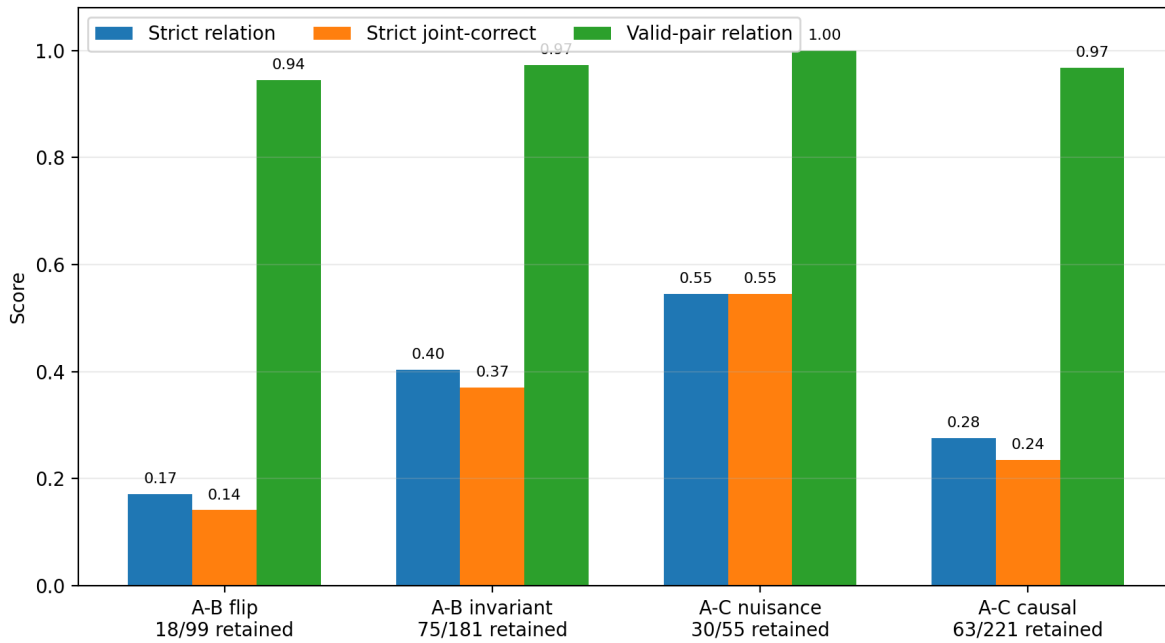


Figure 2. Strict relation, strict joint endpoint-correct, and conditional valid-pair scores. Labels below the axis show retained pairs.

5.4 Structured-output failures and boolean omission

The independent protocol produces 239 terminal format failures. Wrong JSON type for decision accounts for 182 cases (76.2%): the schema requires a string-valued decision label, whereas the model returns a JSON boolean. Free-text values replace canonical booleans in 26 failures and canonical integers in 13. Smaller categories involve datetime, code-set, date-range, threshold, and categorical values. All 1,160 keys ultimately receive terminal records; the final audit classifies every unsuccessful terminal output as a model-format failure.

Table 7. Independent-protocol format failures.

Failure category	Count	Share
Wrong JSON type for decision	182	76.2%
Free text instead of boolean	26	10.9%
Free text instead of integer	13	5.4%
Free text instead of datetime	4	1.7%
Free text instead of code set	4	1.7%
Other code-set failures	3	1.3%
Other categorical failures	3	1.3%
Date-range failures	2	0.8%
Threshold failures	2	0.8%

Boolean tasks reveal the most concentrated failure. Independent strict accuracy is $1/499 = 0.0020$. Among 291 boolean responses that passed the outer response parser, 290 omit the canonical answer and one contains JSON false; no JSON true answer appears. The audit rules out answer inversion and string/boolean normalization errors. In many rows the model supplies a plausible decision, evidence, and explanation while leaving the required answer absent.

Illustrative boolean-output failure
Required: {"answer": false, "decision": "not_a_reimbursement_cap", ...}
Observed: {"decision": "not_a_reimbursement_cap", "evidence": [...], "reasoning": "..."}
Outcome: the observed object omits the required answer field, and strict validation rejects the transaction.

Boolean accuracy rises from 0.0020 under independent inference to 0.8036 under the batched compact protocol.

6. Discussion

6.1 Protocol as a measurement instrument

The 47-point gap makes protocol specification necessary for interpreting benchmark scores. Joint presentation gives the model access to neighboring questions and their shared answer structure, while the compact run requires 80 top-level structured objects instead of 1,160 independent generations. Every reported score is tied to an explicit protocol identifier.

6.2 Content quality and interface reliability

Strict and conditional metrics measure different layers of system performance. Conditional answer accuracy is 0.3583; strict accuracy is 0.2845 after invalid responses return to the denominator. Conditional decision accuracy is 0.6687, while strict decision accuracy is 0.4834. An object without canonical fields is a failed downstream transaction even when surrounding prose contains a plausible conclusion.

6.3 Selection effects in counterfactual reporting

Valid-pair relation scores describe a highly selected subset. A system that returns canonical outputs only for easier pairs can appear almost perfectly stable or sensitive once failures disappear from the denominator. Strict scores estimate complete-system reliability. Joint endpoint-correct scores add a second safeguard by requiring the answers themselves to be right. The three layers together separate availability, relational behavior, and absolute task correctness.

6.4 Relevance for Russian document evaluation

RuDocGround-CF complements MERA with authority conflicts, amendments, temporal scope, cross-document evidence, nuisance interventions, and machine-oriented response contracts. It connects Russian-language evaluation to operational document workflows through exact, grounded, and machine-checkable outputs.

7. Conclusion

RuDocGround-CF adds controlled multi-document reasoning, evidence grounding, structured operational outputs, and cross-variant evaluation to Russian LLM benchmarking. The Qwen3.5 4B baseline reveals a 47-point protocol effect, systematic boolean-answer omission under independent inference, and substantial selection effects in conditional counterfactual scores. Strict, conditional, and joint endpoint-correct metrics jointly measure answer quality, interface reliability, and behavioral consistency.

Data and Ethics Statement

All benchmark documents are controlled synthetic artifacts created specifically for RuDocGround-CF. OpenAI GPT-family models were used as drafting assistants during initial document construction; every document bundle, A/B/C intervention, canonical answer, decision label, and evidence reference was subsequently manually edited and checked, with automated schema, key-alignment, leakage, and content-hash validation. No real company records, confidential materials, or employee personal data were used. The release includes versioned prompts, gold labels, predictions, reports, statistical artifacts, and content hashes.

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Appendix A. Metric Definitions

Strict answer accuracy: exact canonical answers divided by all 1,160 tasks; invalid outputs count as wrong.

Strict decision accuracy: correct decision labels divided by all 693 decision-required tasks; invalid outputs count as wrong.

Evaluatable-output rate: records retained as evaluatable by the released validator, divided by all tasks.

Strict relation score: expected relation satisfied with two evaluator-retained endpoints, divided by all labeled pairs.

Valid-pair relation score: expected relation satisfied, divided by pairs with two evaluator-retained endpoints.

Joint endpoint-correct relation score: both endpoint answers match gold and the expected relation holds, divided by all labeled pairs.

Protocol delta: variant-batched compact metric minus the corresponding independent strict metric.

Appendix B. Results by Answer Type

Table B1. Independent protocol by canonical answer type. Evidence F1 is conditional on evaluator-retained outputs.

Type	Tasks	Evaluator-retained	Format failures	Strict answer acc.	Conditional evidence F1
Boolean	499	291	208	0.0020	0.2757
Categorical	218	215	3	0.1835	0.5933
Code set	52	45	7	0.6346	0.0000
Date	55	55	0	0.5273	0.4558

Type	Tasks	Evaluator-retained	Format failures	Strict answer acc.	Conditional evidence F1
Date range	2	0	2	0.0000	0.0000
Datetime	68	64	4	0.7794	0.8802
Identifier	93	93	0	0.9677	0.9570
Integer	34	21	13	0.2941	0.7619
Money	67	67	0	0.7164	0.6646
Status	70	70	0	0.3714	0.7813
Threshold	2	0	2	0.0000	0.0000

Strict missing-information exact match = 0.7836 [0.7463, 0.8225].

Appendix C. Reproducibility and Artifact Identifiers

Table C1. Frozen identifiers and recorded runtime metadata.

Item	Identifier
Public release reference	v1.3.1
Repository	github.com/tippered1-debug/RuDocGround-CF
Independent predictions SHA-256	08567e35fa0bcc595bd70076494c10e7341ada75d4e76bbbaee256f3ce3f5759
Corrected report SHA-256	9fd3a4558c757f412dcef1c356ab1278ec9cdf625ef8239b0ffb805cda61eff6
Corrected counterfactual SHA-256	d692938edf5b67a0bf0bbebbd27a9a410e94b525f6c33832a4efc5f411f1345c
Bootstrap	30 case clusters; 10,000 replicates; seed 42; percentile 95% CI
Model	qwen3.5:4b; identifier 2a654d98e6fb
Serving	Ollama 0.30.10
Hardware	MacBook Air; Apple M4; 16 GB unified memory

Appendix D. Supplemental Figures

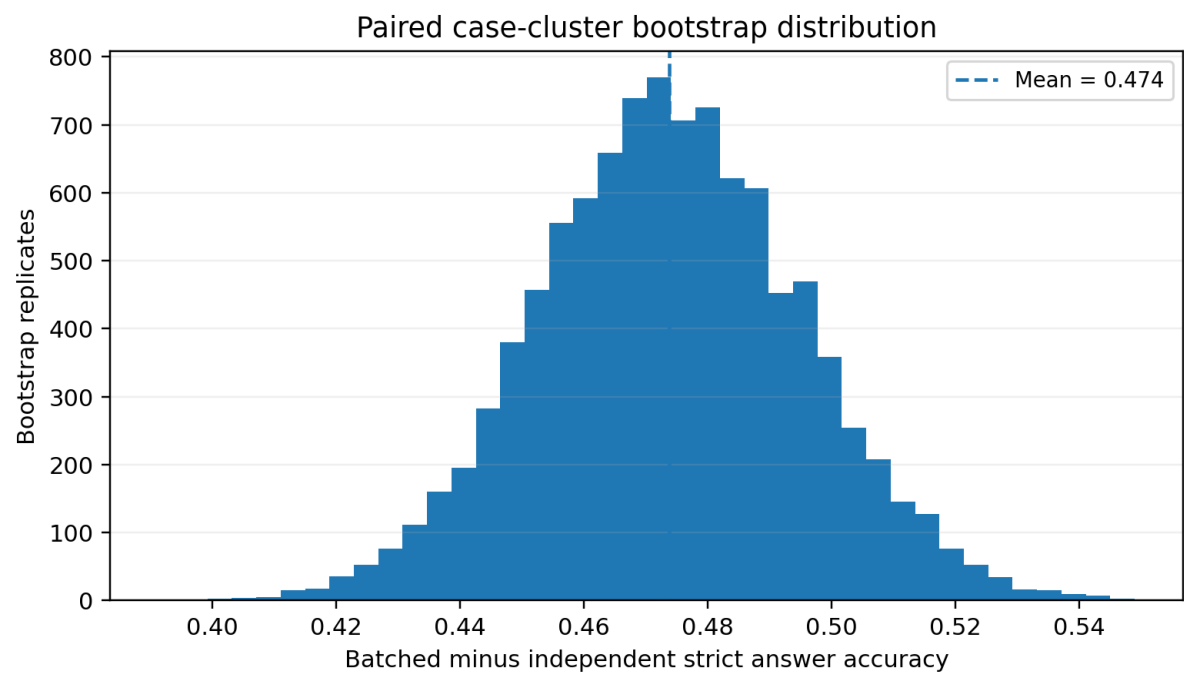


Figure D1. Paired case-cluster bootstrap distribution for the answer-accuracy protocol delta.

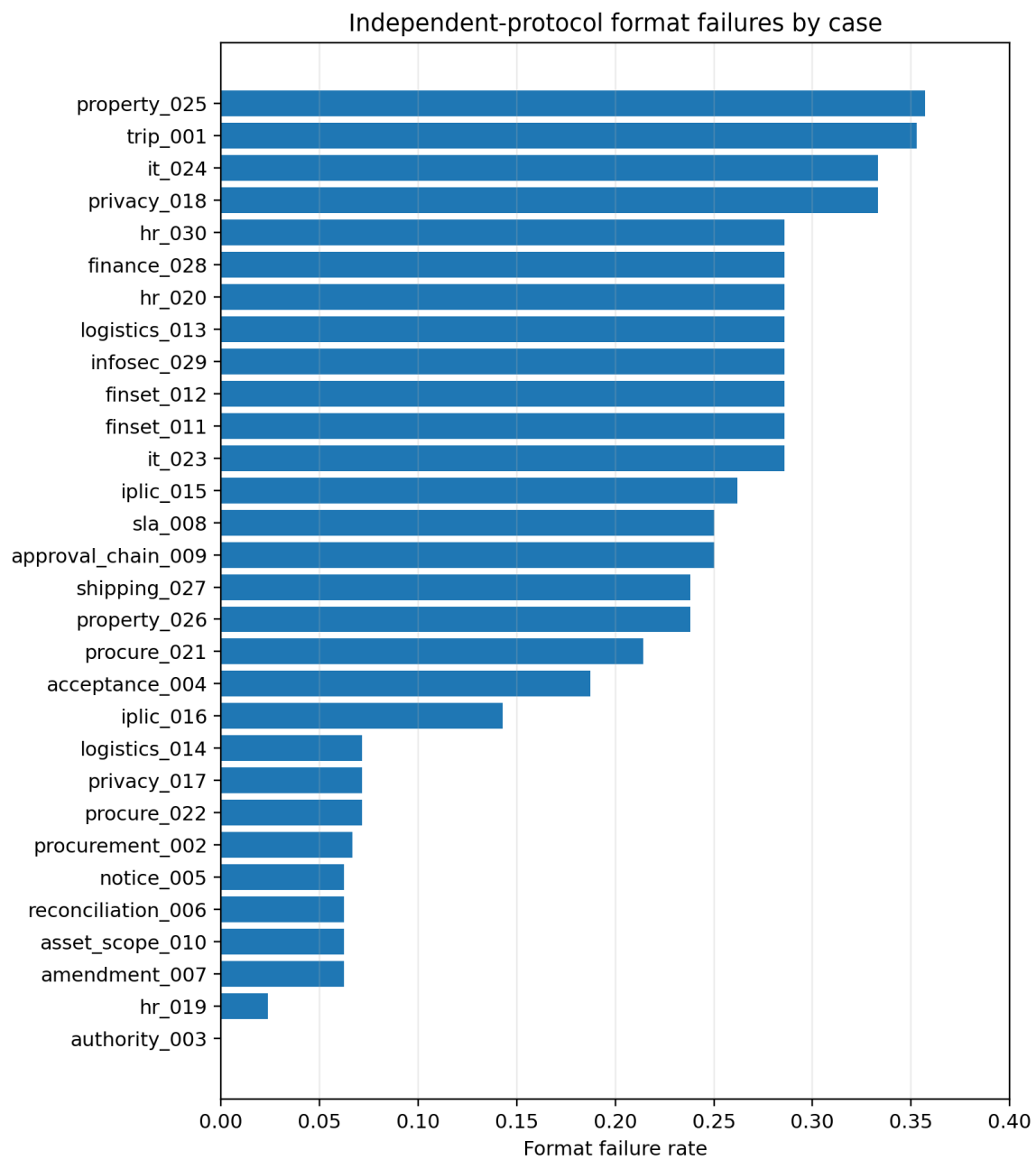


Figure D2. Independent-protocol format-failure rates by case.